

Preliminary

Ver.: 0.03

TFT LCD Specification

Model Name: TD036THEA1

Customer Signature
Date

This technical specification is subjected to change without notice.

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Record of Revision

Rev	Issued Date	Description
0.00	Aug 11, 2004	New create
0.01	Feb 04,2005	Page 5 : 3-1 The pin assignment has been changed to 39 pins from 45 pins. Page 12 : update module drawing. Add insulated tape.
0.02	Apr 18,2005	Page 8 : Add 5.3 TIMING CHART Page 14 : Add 10. PACKING DRAWING

1. FEATURES

The 3.6" LCD module is the active matrix color TFT LCD module. LTPS (Low Temperature Poly Silicon) TFT technology is used and vertical and horizontal drivers are built on the panel. Horizontal scan can be from left to right or from right to left, and Vertical scan can be from up to down or from down to up.

2. GENERAL SPECIFICATIONS

Item	Description	Unit
Display Size (Diagonal)	3.6	Inch
Display Type	Transmissive	-
Active Area (HxV)	72.96 x 54.72	mm
Number of Dots (HxV)	320 x RGB x 240	dot
Dot Pitch (HxV)	0.076 x 0.228	mm
Color Arrangement	RGB Stripe	-
Color Numbers	8 bit RGB (16M color)	-
Outline Dimension (HxVxT) *	80.2 x 66.3 x 3.2	mm
Weight	45	G
Surface Treatment	Hard Coating (3H)	-

* Exclude FPC and protrusions.

3. INPUT/OUTPUT TERMINALS

3.1 TFT LCD Panel

Recommend connector:

Compatible with JAE JAE IL-FHJ-39S-HF-A1

Pin	Symbol	I/O	Description	Remark
1	CP3	C	Capacitor for charge pump	
2	CP4	C	Capacitor for charge pump	
3	CP5	C	Capacitor for charge pump	
4	CP6	C	Capacitor for charge pump	
5	CP7	C	Capacitor for charge pump	
6	CP8	C	Capacitor for charge pump	
7	CPW	C	ONLY FOR WISEPAL	
8	PCDL	C	Capacitor for Pre-charge data signal Low	
9	PCDH	C	Capacitor for Pre-charge data signal High	
10	VCOML	C	Capacitor for VCOM Low	
11	VCOMH	C	Capacitor for VCOM High	
12	AGND	G	Analog ground	
13	PVDD	C	Regulation capacitor for charge pump	
14	AVDD	C	Regulation capacitor for Analog Voltage	
15	CP1	C	Capacitor for charge pump	
16	CP2	C	Capacitor for charge pump	
17	PWM	O	Power transistor gate signal for the boost converter	
18	FB	I	Main boost regulator feedback input	
19	LED-	LED	LED-	
20	LED+	LED	LED+	
21	NC	N	NC	
22	GND	G	Ground	
23	VCC	P	Power supply	
24	VD	I	Vertical sync input	
25	HD	I	Horizontal sync input	
26	DCLK	I	Clock signal, latch data onto line latches at the rising edge	
27	DIN0	I	Data input	
28	DIN1	I	Data input	
29	DIN2	I	Data input	
30	DIN3	I	Data input	
31	DIN4	I	Data input	

32	DIN5	I	Data input	
33	DIN6	I	Data input	
34	DIN7	I	Data input	
35	SDA	I/O	Serial Interface data line	
36	SCL	I	Serial Interface clock line	
37	SCEN	I	Serial Interface chip enable line	
38	SHDB	I	Sleep mode setting pin	
39	GRESTB	I	Global reset pin	

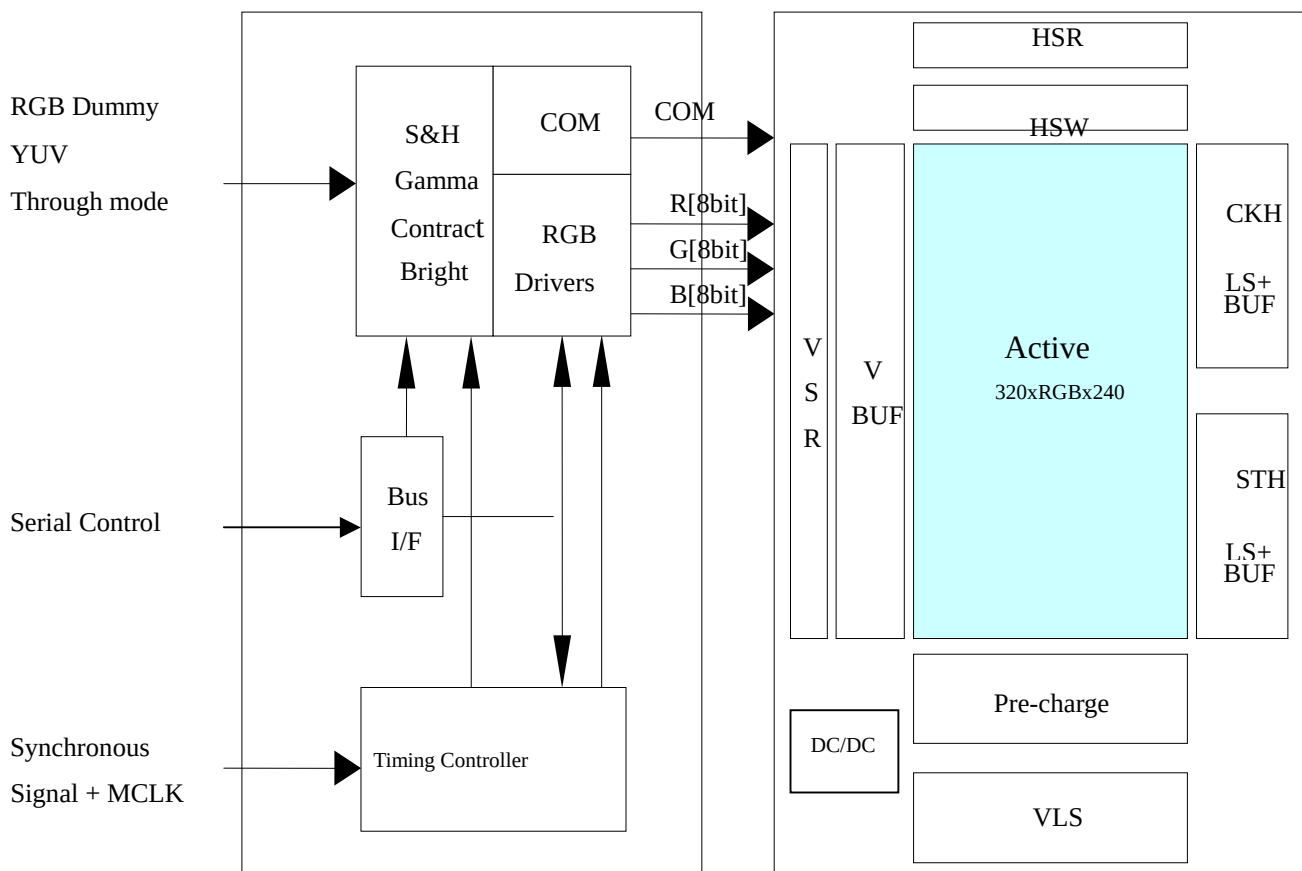
4. ABSOLUTE MAXIMUM RATINGS

VSS=0V

Item	Symbol	Min	MAX	Unit	Remark
Power Supply	VCC	2.7	3.6	V	
Backlight Forward Current	I _F	-	50	mA	
Operating Temperature	Topr	-10	+60	°C	
Storage Temperature	Tstg	-30	+80	°C	

5. ELECTRICAL CHARACTERISTICS

5.1. Driving TFT LCD Panel Block Diagram



5.2. Driving Backlight

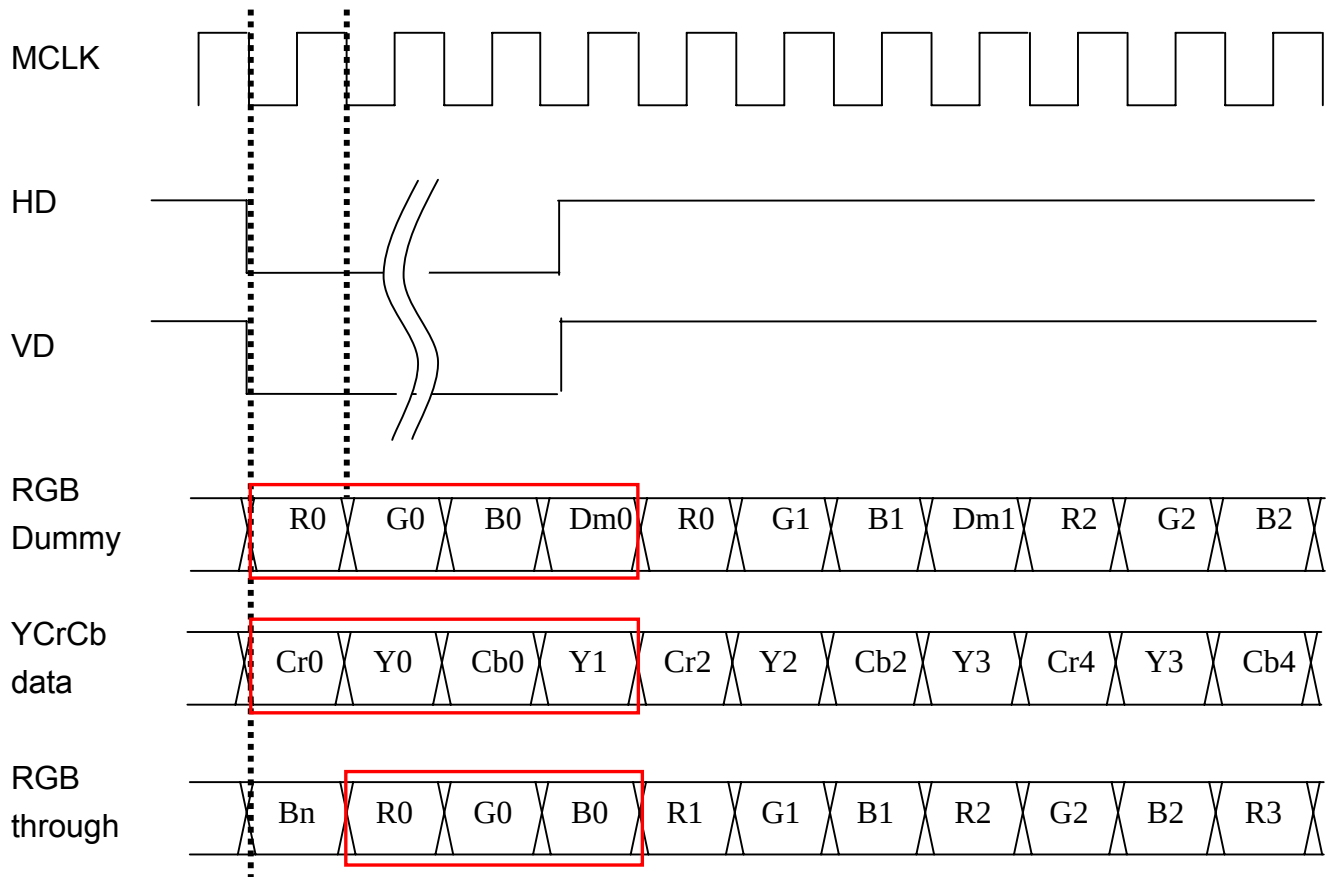
Ta=25°C

Item	Symbol	MIN	TYP	MAX	Unit	Remark
Forward Current	I_F	--	40	46	mA	Note 5-1
Forward Current Voltage	V_{F20}	--	10.8	--	V	
Backlight Power Consumption	W_{BL20}	--	432	--	mW	

Note 5-1: Backlight driving circuit is recommended as the fix current circuit.

5.3TIMING CHART

Input timing (Horizontal + Vertical + Data)



6.OPTICAL CHARACTERISTICS

6.1 Optical Specification

$T_a=25^{\circ}\text{C}$

Item		Symbol	Condition	MIN	TYP	MAX	Unit	Remarks
Viewing Angles		Θ 11	CR $\geq$ 10	30	40	--	Degree	Note 6-1
		Θ 12		30	40	--		
		Θ 21		15	20	--		
		Θ 22		40	50	--		
Contrast Ratio		CR	$\Theta=0^{\circ}$	--	300	--		Note 6-2
Response Time	Rising	Tr	$\Theta=0^{\circ}$	--	13	--	ms	Note 6-3
	Falling	Tf	$\Theta=0^{\circ}$	--	22	--		
Luminance	I _F =40mA	L	$\Theta=0^{\circ}$	--	280	--	cd/m ²	Note 6-4
Chromaticity	White	x _W	$\Theta=0^{\circ}$	--	0.31	--		Note 6-5
		y _W	$\Theta=0^{\circ}$	--	0.34	--		

6.2 Basic Measure Conditions

(1) Driving voltage

Driving voltage with Digital Interface

$V_{CC}=3\text{V}$

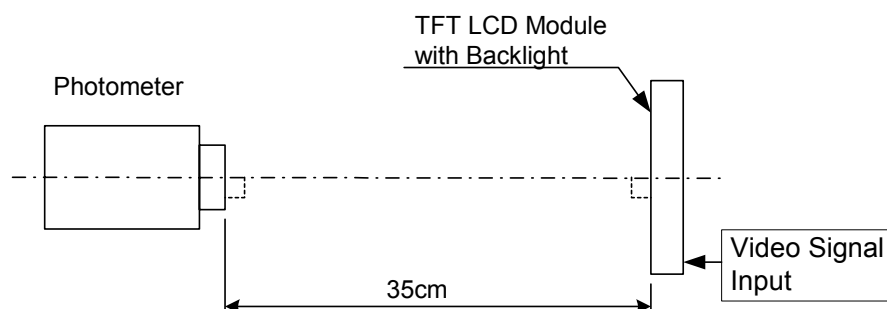
(2) Ambient Temperature: $T_a=25^{\circ}\text{C}$

(3) Testing Point: Measure in the display center point and the test angle $\Theta=0^{\circ}$

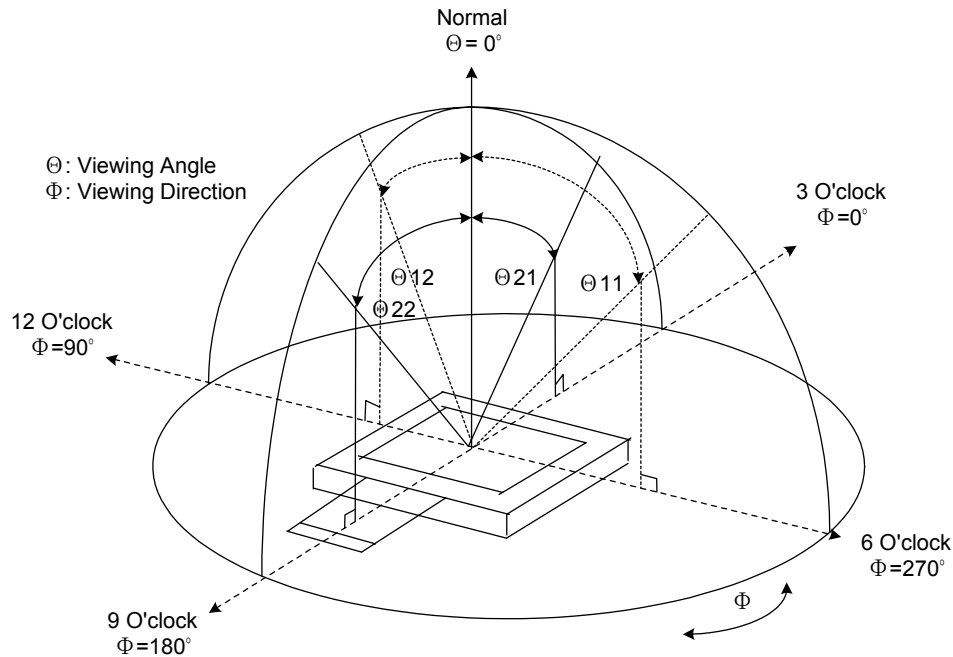
(4) Backlight Current: $I_F=40\text{mA}$.

(5) Testing Facility

Environmental illumination: $\leq 1\text{ Lux}$



Note 6-1: Viewing angle diagrams:

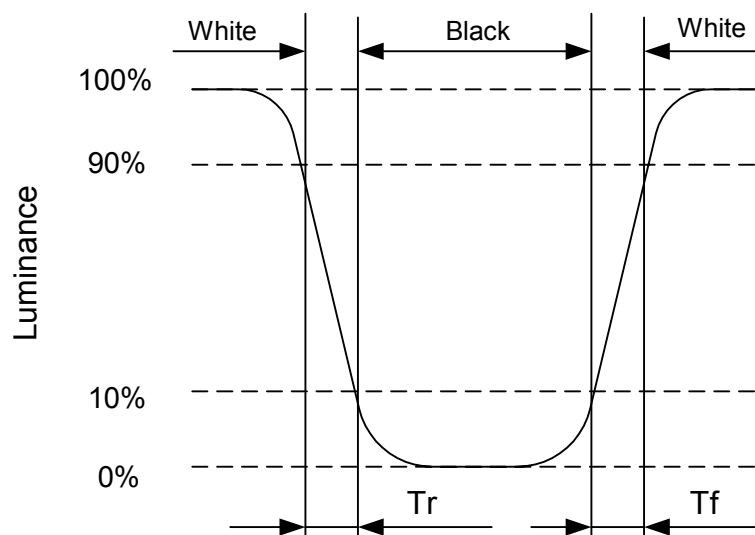


Note 6-2: Contrast Ratio:

Contrast ratio is measured in optimum common electrode voltage.

$$CR = \frac{\text{Luminance with white image}}{\text{Luminance with black image}}$$

Note 6-3: Definition of response time:



Note 6-4: Luminance:

Test Point: Display Center

Note 6-5: Chromaticity: The same test condition as Note 7-4.

7. RELIABILITY

No	Test Item	Condition
1	High Temperature Operation	Ta=+60°C, 240hrs
2	High Temperature & High Humidity Operation	Ta=+40°C, 95% RH, 240hrs
3	Low Temperature Operation	Ta=-10°C, 240hrs
4	High Temperature Storage (non-operation)	Ta=+80°C, 240hrs
5	Low Temperature Storage (non-operation)	Ta=-30°C, 240hrs
6	Thermal Shock (non-operation)	-30°C←→80°C, 50 cycles 30 min 30 min
7	Vibration (non-operation)	Frequency: 10~55Hz; Amplitude: 1.5mm Sweep Time: 11min Test Time: 2 hrs for each direction of X, Y, Z
8	Shock (non-operation)	Acceleration: 100G; Period: 6ms Directions: ±X, ±Y, ±Z; Cycles: Twice

Ta: Ambient Temperature

8 HANDLING CAUTIONS

8.1 ESD (Electrical Static Discharge) Strategy

ESD will cause serious damage of the panel, ESD strategy is very important in handling. Following items are the recommend ESD strategy

- (1) In handling LCD panel, please wear non-charged material gloves. And the conduction ring connect wrist to the earth and the conducting shoes to the earth is necessary.
- (2) The machine and working table for the panel should have ESD prohibition strategy.
- (3) In handling the panel, ionize flowing decrease the charge in the environment is necessary.
- (4) In the process of assembly the module, shield case should connect to the ground.

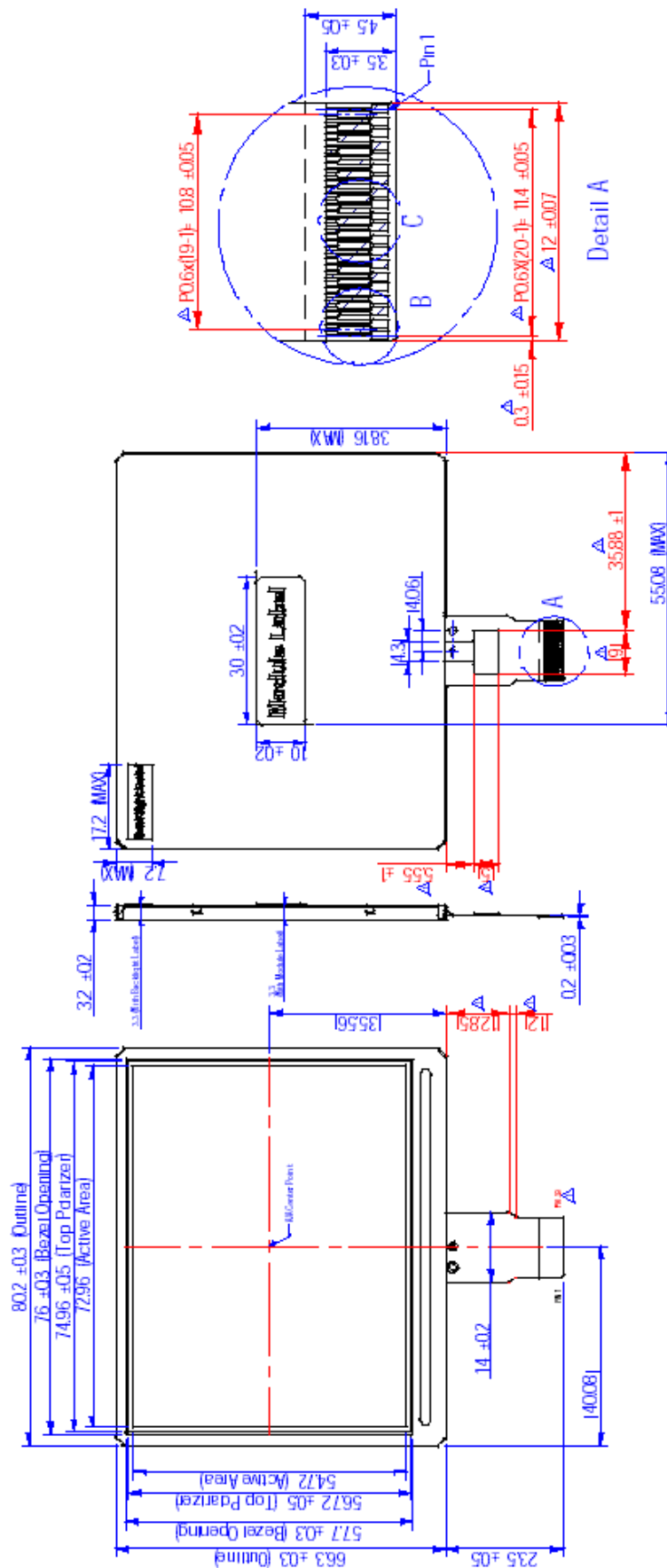
8.2 Environment

- (1) Working environment of the panel should in the clean room.
- (2) The front polarizer is easy damaged, handle it carefully and do not scratch it by sharp material.
- (3) Panel has polarizer protective film in the surface please remove the protection film of polarizer slowly with ionized air to prevent the electrostatic discharge.

8.3 Others

- (1) Turn off the power supply before connecting and disconnecting signal input cable.
- (2) The connection area of FPC and panel is very weak, do not handle panel only by FPC or bend FPC.
- (3) Water drop on the surface or condensation as panel power on will corrode panel electrode.
- (4) As the packing bag open, watch out the environment of the panel storage. High temperature and high humidity environment is prohibited.
- (5) When the TFT LCD module is broken, please watch out whether liquid crystal leaks out or not. If your hand touches liquid crystal, wash your hand cleanly by water and soap as soon as possible.

9 MECHANICAL DRAWING

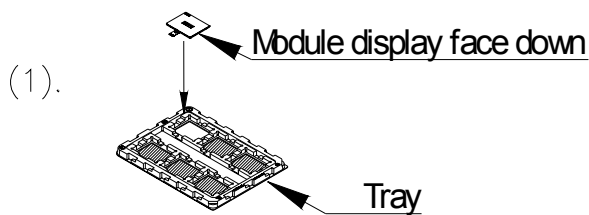


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10 PACKING DRAWING

3.6" module (TD036THEA1) delivery packing method

- (1). Module packed into tray cavity (with Module display face down).
- (2). Tray stacking with 20 layers and with 1 empty tray above the stacking tray unit.
2pcs desiccant put above the empty tray
- (3). Stacking tray unit put into the LDPE bag and fix by adhesive tape.
- (4). Put 1pc cardboard inside the carton bottom, then pack the package unit into the carton.
put 1pc cardboard above the package unit.
- (5). Carton tapping with adhesive tape.



Module Qty on one tray = 6pcs

